# An Efficiency-Uncertainty-Based Consensus Optimization Framework With Dual-Perspective Clustering Analysis for Blockchain Adoption in Supply Chains

Ardavan Babaei, Erfan Babaee Tirkolaee, Sadia Samar Ali, Sankar Kumar Roy, and Gerhard-Wilhelm Weber

***Abstract*—The integration of blockchain technology (BT) in supply chains (SCs) is advancing rapidly due to its benefits in transparency, traceability, security, and decentralization. However, SC managers face challenges in evaluating diverse strategies for effective BT adoption. This study proposes a comprehensive framework combining criteria weighting and evaluation models that prioritize simplicity, idealism, structure, and efficiency. It integrates five decision-making techniques; i.e., simple additive weighted (SAW), entropy method (EM), technique for order of preference by similarity to ideal solution (TOPSIS), super-efficiency data envelopment analysis (SEDEA), and network data envelopment analysis (NDEA), to provide a dynamic, adaptable evaluation process. In addition, a novel risk- and scenario-based optimization model and a predictive comparison approach are introduced to enhance decision-making. The framework is applied to the Norwegian Oil and Gas (O&G) sector, analyzing four BT adoption strategies. It incorporates machine learning (ML) for similarity analysis and dual-perspective clustering to better understand inter-strategy and method relationships. Findings show that a focused, single-purpose blockchain use, especially under high-risk conditions and transformational strategies, is most effective. Key adoption drivers include market and customer pressures, collaboration needs, and limited IT infrastructure. Simplicity and idealism/nonidealism-based models aligned most closely with existing literature, while network structure-based models varied more. The results demonstrate the importance of consensus-based evaluation when selecting blockchain strategies in SCs.**

***Managerial Relevance Statement*—This study provides engineering managers and policy-makers with a practical, robust and evidence-based framework to evaluate and select blockchain implementation strategies in the Oil and Gas Supply Chain (OGSC) sector, even under complex operational conditions and diverse stakeholder demands. By integrating multiple decision-making models and a consensus-based optimization method, the framework enables systematic, comparative, and risk-adjusted strategy assessment. Scenariowise aggregation and dual-perspective cluster analysis using machine learning (ML) clarify similarities among alternatives and evaluation models, supporting informed differentiation and prioritization of strategies. Findings highlight that widely assumed "transformational" strategies are not always optimal; under high-risk conditions, targeted or single-use strategies may deliver greater efficiency gains and risk mitigation. The framework offers a step-by-step operational roadmap from identifying business needs to model-based evaluation, ranking, scenario analysis, and consensus-building, equipping managers and policy-makers with context-sensitive tools to improve supply chain efficiency, transparency, and strategic decision-making.**

***Index Terms*—Blockchain technology (BT), consensus, data envelopment analysis (DEA), decision-making, supply chain management (SCM).**

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## I. Introduction

### A. Blockchain Technology

According to a report by Fortune Business Insights [46], the global blockchain technology (BT) market was valued at USD 27.85 billion in 2024 and is projected to grow significantly, reaching USD 746.41 billion by 2032, with a compound annual growth rate of 49.7%. This substantial growth is driven by increasing digital transformation, rising cybersecurity threats, and expanding applications across industries such as banking, financial services and insurance, healthcare, and supply chain management (SCM). Blockchain is an append-only database that is decentralized, transparent, immutable, resistant to tampering, and operates without the need for trust [1]. BT, characterized

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by a permissioned distributed ledger for storing transaction data, holds promise for the oil and natural gas industry due to its secure data handling within a permissioned network. This industry, reliant on extensive data for operations such as asset history, land management, and financial records, can benefit from blockchain's ability to ensure operations adhere to predefined parameters and securely store process verification data in a distributed ledger [2]. In addition, blockchain offers potential improvements in transparency and accuracy for freight rates, shipment routing, and invoice generation, thereby enhancing SCM, expediting processes, and reducing transactional fraud. Furthermore, blockchain can facilitate the complexities of commodity trading on oil platforms, allowing for the development of trade finance platforms to facilitate crude oil transactions among trading parties. The Oil and Gas (O&G) industry faces significant challenges, including extensive paperwork and reconciliation processes that escalate both monetary and time expenditures associated with transactions. Furthermore, multiparty investment and cooperation in the industry heighten the risks of fraud, errors, and inefficiencies in transactions. Moreover, the costs associated with third-party management are significantly high, along with inefficient negotiations regarding trade and the slow exchange of essential information. Moreover, there is an increased risk of cyber-attacks targeting critical industry data. Consequently, the O&G sector urgently requires a reformation of its management strategies [3].

BT, a relatively novel innovation, has emerged as a promising solution. Since bitcoin's introduction in 2008 [4], BT has evolved markedly over the last ten years. It has moved through various phases: the initial stage dominated by bitcoin, the second phase characterized by Ethereum and self-executing contracts, and the present third stage emphasizing social platforms, all illustrating the development of BT.

### *B. Blockchain Utilization in the O&G and Petroleum Industries*

Despite its broad implementation in diverse sectors, the O&G sector initially approached blockchain cautiously and largely refrained from active involvement. However, with British Petroleum (BP) initiating blockchain trials in 2017, the industry took its primary steps toward integrating BT. Coordinating SCs in the petroleum sector plays a key role in driving global expansion and influence. However, conventional petroleum SC encounters difficulties such as intricate extraction procedures, refining obstacles, distribution challenges, environmental harm, and problems with product traceability. Zhou et al. [5] suggested a blockchain-based product evaluation management system using the interplanetary file system to store evaluation data and smart contracts on Ethereum for retrieval. The system ensured data integrity, reliable retrieval, and sustainability through economic incentives, demonstrating usability, reliability, and sustainability in experimental results.

Advances in technology offer opportunities to transform the petroleum SC, with blockchain and Internet of Things (IoT) sensors enabling greater transparency and security [6]. Müller et al. [7] investigated the Industry 4.0 adoption of small and medium-sized enterprises (SMEs), highlighting challenges in integrating technologies horizontally, vertically, and end-to-end. Using multiple case studies, they identified barriers and enablers, proposing an integration framework. Findings emphasized operational benefits over strategic gains, relying on stakeholder collaboration. Recently, BT has progressed remarkably, becoming a significant trend across multiple industries. Initially approached with caution in the petroleum sector, technologies such as fog computing, IoT, and cloud computing, combined with blockchain, present vast opportunities to improve operational efficiency. For instance, Gazprom Neft, a leading Russian oil company, has demonstrated that digital technologies can enhance productivity in the sector by 10%–15% [8]. The trading of petroleum products such as gasoline and diesel necessitates high levels of security, confidentiality, and efficient data management. However, current systems for managing petroleum trade are often centralized, untransparent, and unreliable. In the petroleum industry, external factors greatly affect SCM, highlighting the need for a thorough understanding and ongoing improvement efforts [9]. Given the crucial role of the petroleum sector in the global economy to satisfy high-volume energy demands, digitalization is becoming progressively necessary in all stages, including extraction, refining, marketing, and distribution.

In the dynamic context of petroleum SCM, ensuring product integrity during transport presents inherent risks in traditional centralized systems. However, the rise of BT marks a significant transformation, rendering a decentralized approach that enhances transparency and accountability [10]. This innovative technology addresses the intricacies of global SCs, paving the way for improved transparency, traceability, and security [11]. Researchers have been also investigating blockchain's diverse applications in SCM and sustainability projects, and overcoming implementation challenges, which in turn are revolutionizing digital currency functions and alleviating concerns related to cost, storage, and security [12], [13]. As a form of distributed ledger technology (DLT), blockchain consists of a sophisticated network of linked, time-stamped records safeguarded by cryptographic techniques. Its decentralized structure enables consensus and validation across the network of nodes, ensuring a secure, unchangeable, and trustless transaction framework [14]. Beyond traditional business functions, blockchain eliminates the need for intermediaries, acting as a distributed database. Known for maintaining transactional confidentiality, integrity, and availability, blockchain functions as a shared, transparent ledger across a peer-to-peer network. Data and transactions are securely recorded and verified cryptographically, ensuring immutability once added and requiring majority consensus for any changes [15].

In summary, the O&G industry is increasingly recognizing the value of BT, particularly when integrated with advanced technologies such as IoT and AI. This study explores the application of BT in the petroleum sector, focusing on its role in decision-making and SC optimization. By addressing existing gaps and introducing innovative frameworks, the aim is to provide actionable insights for industry leaders, laying the foundations for a more efficient, secure, and sustainable O&G industry.

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### C. Research Questions

*RQ1:* How can blockchain strategies be effectively evaluated for the OGSC despite its complexity and limited prior research?

*RQ2:* What comprehensive evaluation framework can be developed to investigate blockchain deployment strategies in the O&G sector, incorporating multiple decision-making philosophies?

*RQ3:* How can evaluation results from different decision-making methodologies be consolidated to provide an optimal unified perspective for OGSC managers?

*RQ4:* What policy implications, particularly risk, statistics, and scenario-based approaches, should be considered when aggregating evaluation results for BT adoption in the O&G sector?

*RQ5:* How do novel concepts (e.g., prediction-based approaches for comparing decision-making methods) enhance the evaluation of BT adoption strategies compared to traditional methods?

### D. Research Objectives

1) *Evaluating Blockchain Deployment Strategies:* This research assesses various BT adoption strategies for using blockchain in the O&G industry to understand their efficacy and performance,
2) *Employing Different Decision-Making Philosophies:* We apply network decision-making and data-driven models based on simplicity, idealism, and nonidealism philosophies, and super-efficiency concepts to evaluate blockchain deployment strategies comprehensively,
3) *Developing a Consolidation Method:* An optimization using mathematical programming techniques is developed to consolidate evaluation results obtained from different decision-making approaches to provide SC managers with a unified perspective,
4) *Addressing Policy Implications:* We finally investigate and present policies for SC managers regarding the aggregation of evaluation results, particularly focusing on risk and scenario-based approaches. This study, for the first time compared to the literature, indicates the importance of consensus from a statistical perspective on prediction, contrasting it with traditional approaches.

### E. Research Structure

The rest of this article is organized as follows. The literature is reviewed in Section II to highlight the main contributions. The problem is stated in Section III, where the evaluation models are described in Section IV. Section V represents the obtained results, which are then discussed in Section VI. Finally, Section VII concludes the article and offers a useful outlook for future studies.

## II. Literature Review

Here, the research works addressing the adoption BT in SCs with the help of multiattribute decision-making and mathematical optimization methods are critically reviewed. As an important point, efforts to integrate BT into optimization and data-driven models in the petroleum sector are still in the early stages. Babaei et al. [16] introduced a dual-objective optimization model incorporating blockchain's clarity into a sophisticated three-level SCM system. Using various algorithms, they aimed to evaluate essential factors such as expenses, service quality, and environmental impact [17], [18]. To meet the industry's evolving demands, Babaei et al. [19] offered an ecofriendly circular SCM system enhanced with blockchain transparency. In addition, multiobjective optimization frameworks were also developed to address challenges related to cost-effectiveness, traffic congestion, and openness [20]. Sharma et al. [21] investigated BT adoption in the agriculture SC, comparing trends in developed and developing economies. Using data from The Netherlands, United States, Saudi Arabia, and India, they specified enabling factors and barriers, highlighting policies as the key enabler.

Cutting-edge research investigates synchromodal transportation strategies combined with BT, aiming to align with environmental sustainability goals. Oudani et al. [22] proposed an environmentally friendly blockchain model aimed at improving energy efficiency in the management of hazardous materials within SCs. Similarly, Carvalho et al. [23] crafted an extensive model dedicated to refining strategies for blockchain integration and the design of SCM systems. Munim et al. [24] applied a strategic decision-making model to assess BT adoption within the oil industry, fostering novel methodologies. Babaei et al. [25] emphasized the significant impact of risk and uncertainty on decision-making regarding new technologies in SCs. Their study examined how blockchain strategies can mitigate these challenges, emphasizing optimization models tailored to manage SC dynamics and technological uncertainties, but it does not integrate decision-making methods or consolidate evaluation results. Yousefi and Tosarkani [26] performed a thorough study to pinpoint and rank the factors that facilitate blockchain implementation in the mineral SC industry. Lotfi et al. [27] explored the adoption of resilient and sustainable SC network designs powered by BT. Kumar et al. [28] examined the significance of Industry 4.0 in revolutionizing industries, emphasizing the digital food SCM system's role. They suggested some frameworks to evaluate system performance, optimizing blockchain-specific metrics such as latency, throughput, and round-trip time for improved outcomes. Furthermore, Goli [29] utilized a mathematical approach to enhance the design of product portfolios and manage financial transactions within circular SCM systems by adopting BT. Similarly, Rahmanzadeh et al. [30] introduced a strategic model for SC planning that incorporates open innovation. They employed a fuzzy logic method to refine decision-making processes in alignment with both SC goals and principles of open innovation. They also suggested a blockchain-driven registration system to tackle intellectual property issues.

Recently, Abdel-Basset et al. [31] developed a multi-criteria decision-making (MCDM) framework based on the analytical hierarchy process and technique for order of preference by similarity to ideal solution (TOPSIS) to assess the effect of technologies of Industry 5.0. It was revealed that blockchain, 5G, drones, nano-technology, Big Data, robotics, IoT, cloud computing, and artificial intelligence (AI) play key roles.

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Ye et al. [32] investigated the adoption of BT in agri-food SCs, evaluating benefits and costs. They explored conditions under which competing SCs benefit from BT adoption and proposed optimal government subsidy schemes. The obtained results highlighted competitive dynamics and the need for subsidies to enhance consumer surplus and social welfare. Gardas et al. [33] presented a novel node selection procedure for blockchain-enabled edge IoT systems, addressing cloud-based IoT issues. Using fuzzy logic and TOPSIS, their framework enhanced node selection, improving response times, resource allocation, and data trust. Büyüközkan and Tüfekci [34] applied a decision-making process using the VIseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR) method to appraise enterprise blockchain platforms. It combined multiple preference formats, aiding organizations in selecting the most suitable blockchain platform, thus enhancing blockchain literature. They concluded that as blockchain platforms are increasingly utilized in diverse business applications, evaluating them becomes crucial. Lai and Liao [35] offered a hybrid methodology based on linguistic D numbers, double normalization-based multiple aggregation, and criteria importance through intercriteria correlation technique to address information uncertainty and criteria correlations. They evaluated information uncertainty, variety of criteria, and the correlations between criteria in a numerical example. Filatovas et al. [36] suggested a framework for choosing suitable consensus protocols for DLTs using MCDM techniques. They addressed varying requirements, demonstrating their effectiveness with common blockchain systems and bike-renting applications. Erol et al. [37] emphasized the growing importance of sustainable development in building resilient SCs. Their proposed approach was based on an integrated fuzzy MCDM model to assess blockchain's applicability in sustainable SCs, identifying sourcing, delivery, transformation, and product recovery as the most suitable functions. Dehshiri et al. [38] addressed renewable energy development challenges by proposing blockchain strategies to mitigate risks. With the help of strengths, weaknesses, opportunities, and threats analysis and gray MCDM methods, they demonstrated blockchain's potential to decentralize the energy system, reducing renewable energy production costs and prices. Kara et al. [39] developed a decision support system for selecting blockchain platforms for digital projects. Employing MCDM and interval-valued spherical fuzzy sets, they examined a case study along with an extensive sensitivity analysis. Babaei et al. [47] proposed a decision support framework to evaluate blockchain deployment in the OGSC, focusing on cost, profit, and income efficiency using DEA models. Applied to Norway's O&G sector, the results showed varying effectiveness of strategies under deterministic and uncertain conditions. The literature review highlights the growing interest in BT adoption across various industries, including SCs. Table D.1 in the Supplementary Material summarizes key contributions from the literature, identifies research gaps, and outlines how our study advances the field.

To address the critical need for a robust research methodology, our study presents and applies a comprehensive, multifaceted approach to evaluating BT adoption strategies within SCs. The methodology is based on a combination of established decision-making models; i.e., entropy method (EM), TOPSIS, simple additive weighted (SAW), and data envelopment analysis (DEA) models, each selected for its distinctive ability to offer insightful and complementary perspectives on the data. By integrating these models, our approach facilitates a thorough evaluation that balances simplicity and efficiency while providing a holistic view. The EM, for instance, objectively determines the importance of criteria based on data variability, while TOPSIS ranks options by their proximity to an ideal solution. SAW offers a straightforward approach for initial assessments, and DEA conducts an in-depth efficiency analysis across multiple inputs and outputs. This layered methodology not only strengthens the comprehensiveness of our findings but also enables a comprehensive understanding of blockchain implementation strategies within SCs. Our findings are further validated through practical application in the Norwegian O&G sector, demonstrating the efficacy of our methodology in real-world scenarios. In accordance with the literature review and introduction, the research contributions can be summarized as follows.

### A. Contributions

1) *Comprehensive Evaluation:* This study delivers an in-depth assessment of various BT adoption strategies within the petroleum sector. By employing multiple decision-making approaches, including SEDEA, SAW, TOPSIS, and NDEA, we offer a comprehensive understanding of how different strategies perform. This approach allows for a thorough evaluation of blockchain's impact on OGSC optimization, addressing gaps identified in previous research, such as those by Babaei et al. [16], [47] and Munim et al. [24], who explored blockchain integration but did not fully consolidate the results across various methodologies.
2) *Utilization of Various Decision-Making:* Our methodology integrates diverse decision-making techniques, each rooted in different theoretical philosophies; i.e., network decision-making, data-driven models, and consensus optimization frameworks, to assess blockchain deployment strategies. This multifaceted approach contributes to a more robust evaluation framework, enhancing the understanding of blockchain's role in improving SC efficiency and addressing complexities that were previously studied in isolation (e.g., [19], [34], [35]).
3) *Consolidation of Evaluation Results:* Introducing risk and scenario-based policies to consolidate evaluation results from different methodologies offers a novel approach to address the need for comprehensive insights for OGSC managers. By consolidating diverse results into a cohesive, mathematically optimized risk-scenario-based framework, we address the need for holistic insights that previous studies, such as those by Babaei et al. [25], have highlighted but not fully realized. In addition to tackling optimization challenges, this article adopts a dual-perspective clustering analysis informed by machine learning (ML) to uncover similarities among strategies and relationships between evaluation methods. Especially, the dual perspective clustering approach can enable both

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understanding the relative position and similarities between the different blockchain strategies, and shedding light on the methodological similarities among the evaluation models.

4) *Implications:* This research offers strategic and operational insights into BT adoption and the evaluation of BT adoption strategies. In addition, it introduces novel statistically-grounded concepts, including a prediction-based framework for method comparison. In addition, this research presents analyses of the application of blockchain strategies in the O&G industry by integrating ML approaches with optimization and decision-making models. These approaches were overlooked by Babaei et al. [25] and Munim et al. [24].

### B. Contributions to Engineering Management Practice and Literature

This research is highly relevant to engineering management and the academic literature, particularly in areas such as technology adoption, supply chain optimization, and planning under uncertainty. In capital-intensive sectors such as O&G, managers face complex dilemmas with emerging technologies such as blockchain. This study provides a robust, multimethod framework (SAW, EM-TOPSIS, SEDEA, NDEA) to compare blockchain implementation strategies, single-use, localization, substitution, and transformation, addressing vagueness and conflicting outputs common in single-method approaches. By introducing a risk-scenario-based consensus optimization model, the framework incorporates uncertainties and varying risk appetites, guiding managers to select the most appropriate strategy under real-world conditions. The consensus-making approach resolves conflicts across evaluation philosophies, supporting stakeholder alignment and confident strategic choices.

In addition, a stepwise operational roadmap, from identifying needs and criteria to model-based evaluation, scenario analysis, consensus-building, and prioritization, provides practical guidance. Dual-perspective clustering (using ML techniques) offers nuanced insights into strategy and method similarities, enabling a deeper understanding of strategic landscapes. The integration of diverse decision-making philosophies within a single framework, along with a validated consensus optimization model, fills a significant gap in the literature by aggregating heterogeneous evaluation outcomes into rigorous, actionable engineering management decisions.

## III. Problem Description

Adopting blockchain in the OGSC is crucial due to its ability to address significant challenges, including complex extraction processes, refining, distribution, environmental impact, and traceability issues. Blockchain, combined with IoT sensors, enhances transparency, security, and efficiency by offering a decentralized, immutable ledger system. Blockchain ensures the integrity and confidentiality of petroleum products such as gasoline and diesel, which demand high security. This technology streamlines operations, reduces costs, and eliminates the need for intermediaries by providing a secure and transparent ledger. The decentralized nature of blockchain enhances accountability and traceability, addressing the vulnerabilities of centralized systems. Thus, BT adoption is vital for improving operational efficiency and maintaining product integrity in the global petroleum SC. BT adoption also involves alternative strategies. However, adopting blockchain presents some challenges. In this work, we investigate which strategy is most suitable for the O&G industry. The focus is on decision-making, efficiency analysis, and the consensus procedure. In other words, the presence of sensitive data, involvement of multiple stakeholders, extensive paperwork, and the complexity of transactions in the O&G industry underscore the urgent need for adopting BT within its SC. In response, SC managers must diligently assess blockchain deployment strategies using comprehensive criteria. Subsequently, aggregating the evaluation results becomes crucial to make a well-informed decision. Employing an aggregation method based on the scenario approach allows for the incorporation of uncertainty and risk factors into the decision-making and evaluation processes of SC managers. As such, this section is segmented into four sections, each representing distinct aspects of this approach.

A pseudocode flowchart outlining the research methodology steps and scope is provided in Figure E.1 in the Supplementary Material. In the remainder of this section, we describe the blockchain strategies and the criteria needed to evaluate them. Next, the evaluation models are outlined based on decision-making concepts. Finally, we discuss the consensus process for reconciling results from different decision-making models.

### A. Blockchain Adoption Strategies

BT adoption can be divided into four stages based on these dimensions: Single-use, localization, substitution, and transformation. Munim et al. [24] offered these four strategic approaches, delineated by the interplay of complexity and novelty. First, the "single-use" strategy entails minimal novelty and coordination, rendering it ideal for small-scale experimentation. In contrast, the "localization" strategy embraces high levels of novelty but requires relatively low coordination, facilitating streamlined operations within localized networks while augmenting transparency. The third strategy, termed "substitution," necessitates considerable coordination and relies upon established applications, albeit with cautionary advisories regarding potential challenges. Finally, the "transformation" strategy mandates heightened levels of coordination and novelty, denoting a significant paradigm shift within systems. As depicted in Fig. 1, these blockchain strategies provide a comprehensive framework for navigating the complexities of SC integration with BT.

### B. Criteria for Assessing Blockchain Adoption Strategies

According to Munim et al. [24], the adoption of BT within SCs is influenced by a range of factors, both motivating and inhibiting. Several factors drive the adoption of BT in industries such as O&G. These include the necessity for collaborative efforts to optimize technology use, consumer demand for sustainable products driving transparency and credibility in SCs, and market

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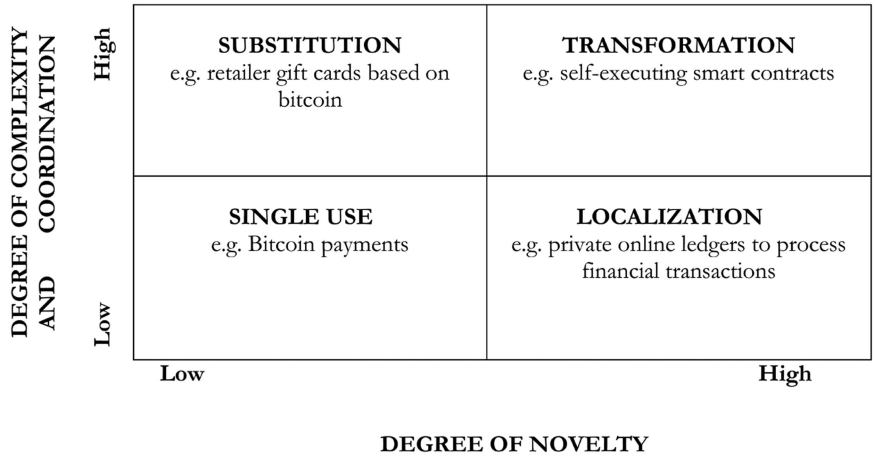


Fig. 1. Blockchain adoption strategies [40].

pressures pushing organizations toward BT adoption to meet sustainability targets and market expectations.

Moreover, three main factors contribute to the integration of blockchain: Enhancing security, reducing operational costs, and improving traceability. Blockchain's ability to track and trace products improves transparency and traceability, offering a competitive edge and meeting consumer expectations for sustainable products. However, several obstacles stand in the way of effective blockchain adoption. These include organizational challenges such as lack of technological expertise, inadequate implementation tools, and a lack of benchmarking data. SC-related obstacles include gaps in customer awareness, reluctance among partners to share information, and coordination challenges. In addition, technological barriers such as technology immaturity, infrastructure limitations, and security concerns pose significant challenges to blockchain adoption.

In summary, according to Munim et al. [24], the adoption of BT in SCs is influenced by fifteen factors. Drivers include collaboration with SC partners (NCS), consumer demand for sustainable products or customer pressures (CP), market pressures (MP), enhanced security (IIS), reduced operational costs (ROC), and improved traceability (IIT). Barriers comprise a lack of technological expertise (LE), inadequate implementation tools (LT), insufficient benchmarking data (LB), gaps in customer awareness (LCA), reluctance among SC partners to share information (LSCB), coordination challenges (LSCO), technology immaturity (TI), limited IT infrastructure (LI), and security concerns (SC). In the DEA model, outputs represent the desired outcomes (drivers) to be maximized, while the remaining criteria are treated as inputs. In the network-based DEA model, NCS (Need for Collaboration with SC Partners), CP (Customer Pressures), and MP are classified as intermediate inputs. This classification is due to the fact that these factors arise from the influence of other inputs and outputs, and reflect the pressures an organization recognizes and evaluates during the assessment process. In other words, by identifying the other inputs and outputs, the organization is able to recognize the intermediate inputs, which are essentially pressure-related factors.

Understanding these fifteen factors is crucial for effectively navigating the complexities of BT adoption strategies, enabling organizations to harness its transformative potential while mitigating associated challenges.

## IV. Evaluation Models for Assessing Blockchain Strategies

Various evaluation models are grounded in distinct decision-making logics and philosophies. To attain a comprehensive decision on the implementation of an efficient blockchain strategy, SC managers must evaluate strategies from various perspectives, enabling them to decide on BT adoption from a holistic standpoint. Consequently, a consensus model is built up with the help of several commonly utilized methods that incorporate various decision logics. The details are given in the Supplementary Material (Section A).

### A. Consensus Model

This section focuses on reconciling evaluation outcomes derived from diverse decision-making models such as network and SEDEA, TOPSIS, and SAW, each embodying distinct decision-making philosophies. Achieving robust and accountable decisions for SC managers necessitates merging the rankings and outcomes provided by these models. One method of aggregation involves minimizing discrepancies among rankings. To quantify such discrepancies, the Manhattan distance function measures the difference between each strategy's rank and every conceivable rank, considering the total number of strategies. The summation of intervals between possible ranks and the rank derived from the evaluation method for each strategy signifies the level of disagreement [41]. This aggregation challenge is akin to an assignment problem, where the objective function seeks to minimize discrepancies while assigning each strategy to a rank and each rank to a strategy. To address this, we introduce scenario-based models. Scenarios are formulated based on disagreement values, particularly when these values are closely aligned. For instance, if the discrepancy between the values of two strategies is minimal, introducing a mutation between them introduces randomness. This mutation accounts for uncertainties and unconscious decision-making biases, facilitating a more

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TABLE I
Evaluation Results of the Blockchain Deployment Strategies Using Various Analytical Models

| Strategies | SEDEA | SAW | TOPSIS | NDEA | Munim et al. [24] |
|---|---|---|---|---|---|
| Single-use | 1.537 | 0.600 | 0.948 | 1.451 | 0.706 |
| Localization | 1.154 | 0.512 | 0.469 | 0.937 | 0.758 |
| Substitution | 1.131 | 0.436 | 0.088 | 0.949 | 0.858 |
| Transformation | 1.063 | 0.426 | 0.058 | 0.871 | 0.898 |

comprehensive exploration of the solution space. We delineate two approaches:

*Approach I.* It formulates an objective function based on the weighted sum of scenarios. *Approach II.* It prioritizes minimizing the maximum deviation (risk).

Objective function (1) incorporates both approaches, while (2) and (3) address the assignment problem, with constraint (4) defining decision variables. It is noteworthy that the model presented in (1)–(4) is an assignment optimization model[1] within the field of mathematical programming. Accordingly, decision variables are defined for assigning each rank to the respective blockchain strategy.

$$\text{minimize} \quad \begin{cases} I : \sum_e \sum_d b_\omega g_{ed}^\omega \, h_{ed} \\ II : \rho \\ \text{such that} \sum_e \sum_d g_{ed}^\omega \, h_{ed} \leq \rho \end{cases} \tag{1}$$

$$\text{subject to} \quad \sum_e h_{ed} = 1 \qquad \forall d, \tag{2}$$

$$\sum_d h_{ed} = 1 \qquad \forall e, \tag{3}$$

$$h_{ed} \in \{0, 1\}, \rho \text{ free variable} \qquad \forall e, d. \tag{4}$$

The results of this assignment are outlined in Table C.1 in the Supplementary Material for both approaches. Here, we utilize three distinct decision models, each on the basis of specific philosophical approaches, to evaluate BT adoption strategies. These models include the EM-TOPSIS method, which emphasizes criteria with higher variability and distances from anti-ideal points; the DEA method, which leverages network analysis and super-efficiency from data-driven optimization; and the SAW method, known for its simplicity in evaluation. These approaches form the foundation for assessing blockchain implementation strategies. The results from these methods are then analyzed using a risk-based optimization model and consensus scenario. Fig. 2 provides a visual representation of our evaluation framework.

## V. Results

In this section, we focus on the evaluation results obtained from the case study problem discussed in the Supplementary Material (see Section B). The evaluation process includes the assessment of the criteria (see Fig. 3), determination of the evaluation results using decision-making models (see Table I), identification of the differences across these models (see Fig. 4), examination of the level of consensus among different evaluation outcomes (see Table II and Fig. 5), and comparison of our results with those of the baseline study from which the data were sourced (see Fig. 4 and Table III). We also justified incorporating scenarios to support consensus, as presented in Table C.1 in the Supplementary Material. Initially, the weights of criteria were determined according to Equations (A1)-(A5) in the Supplementary Material (Section A), as illustrated in Fig. 3.

These criteria comprises various aspects of the BT implementation context, including the need for collaborating with SC partners (NCS), customer pressures (CP), MP, enhanced security (IIS), ROC, improved traceability (IIT), lack of expertise about

[1]For more details, please refer to Armstrong et al. [42].

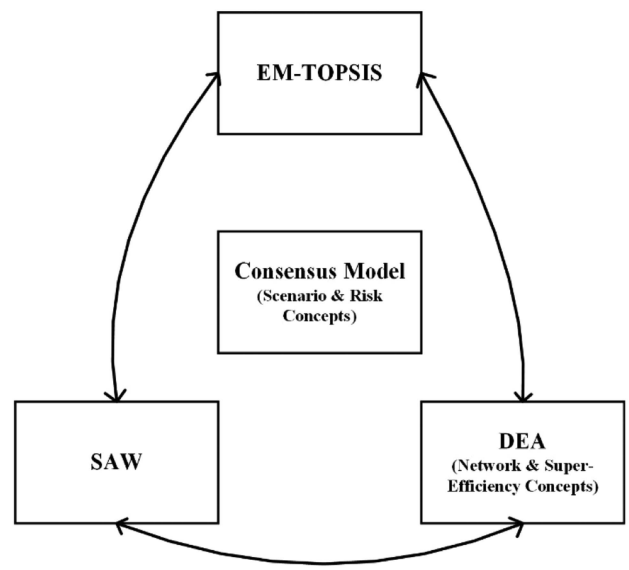


Fig. 2. Our proposed framework.

TABLE II
Disagreement Values for the Blockchain Deployment Strategy Rankings

| Strategies | Position 1 | Position 2 | Position 3 | Position 4 |
|---|---|---|---|---|
| Single-use | 3 | 6 | 9 | 12 |
| Localization | 7 | 2 | 3 | 8 |
| Substitution | 8 | 3 | 2 | 7 |
| Transformation | 9 | 7 | 5 | 3 |

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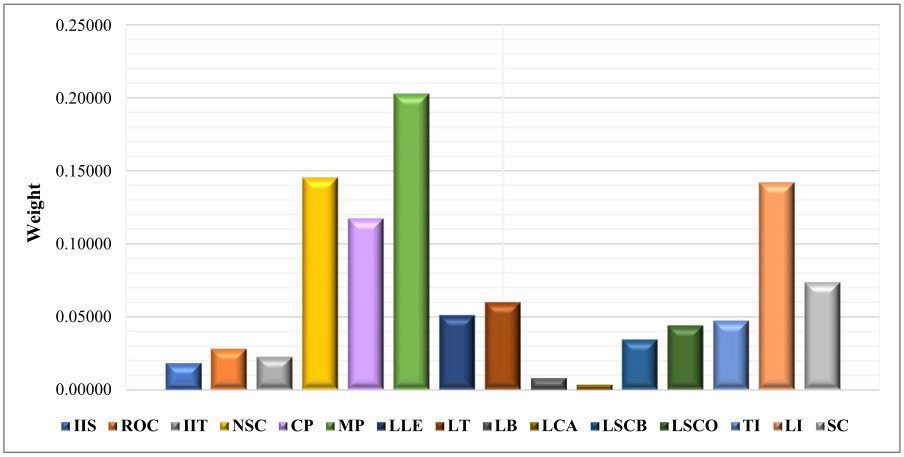


Fig. 3. Weight distribution of the criteria.

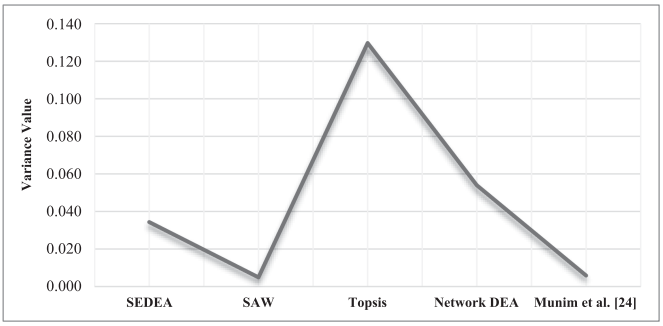


Fig. 4. Dispersion of the scores across evaluation models.

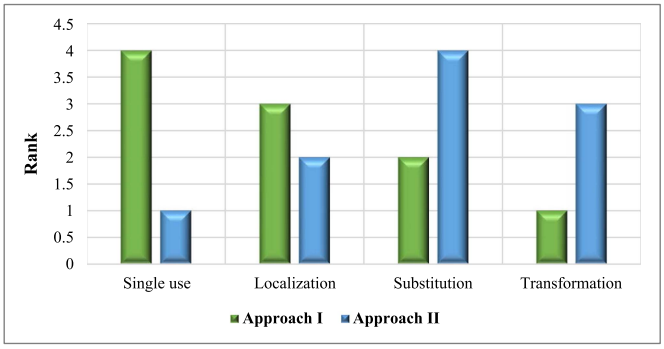


Fig. 5. Consolidated evaluation results for the blockchain deployment strategies.

technology (LE), lack of tools for BT implementation (LT), lack of benchmarking data for BT implementation (LB), lack of customer awareness about BT (LCA), lack of SC partner collaboration (LSCB), lack of SC partner coordination (LSCO), technology immaturity (TI), limited information technology infrastructure (LI), and security concerns (SC). Among these criteria, MP, NCS, LI, and CP emerged with the highest values in terms of weight, respectively. This indicates their paramount importance within the context of BT implementation.

TABLE III
Investigation of $R^2$ and Adjusted $R^2$

| Case Study | Our Models | | | | |
|---|---|---|---|---|---|
| | Indexes | SEDEA | SAW | TOPSIS | NDEA |
| Munim et al. [24] | $R^2$ | 0.702592 | 0.906146 | 0.919549 | 0.627929 |
| | Adjusted $R^2$ | 0.553888 | 0.937431 | 0.879323 | 0.441893 |

Table I displays the results of assessing blockchain deployment strategies. These results comprise the outputs of (A14)–(A17) utilizing SEDEA, (A13) employing SAW, (A6)–(A12) under the framework of TOPSIS, and (A18)–(A22) within the context of NDEA, alongside findings from Munim et al. [24]. In the super-efficiency model, SAW, and TOPSIS analyses, single-use and localization strategies achieved Ranks 1 and 2. Conversely, within the NDEA approach, single-use and substitution strategies secured Ranks 1 and 2. In Munim et al. [24], transformation and substitution strategies attained Ranks 1 and 2. Such divergent outcomes may lead to confusion among SC managers, emphasizing the necessity for consensus in evaluating these results.

Our study utilized four decision-making methods to evaluate BT adoption strategies and compared these results with the findings of Munim et al. [24]. The comparison is illustrated in Fig. 4 and Table II. The dispersion of scores acquired through diverse evaluation models is delineated in Fig. 4. Increased dispersion in scores implies enhanced differentiation ability for decision-makers in selecting superior and more fitting strategies. As shown, the results from Munim et al. [24] showed the lowest ability to distinguish between strategies, similar to the basic SAW method. In contrast, our proposed decision-making approaches display significantly stronger differentiation power. The disagreement values in evaluation outcomes are documented in Table II. The results highlight that the "single-use" strategy ranks first, followed by the localization strategy in second place, with the transformation strategy securing the third and fourth ranks, showcasing minimal disagreement values.

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Regarding alterations, since Munim et al. [24] identified the transformation strategy as the optimal choice, we interchange the disagreement values between the "single-use" and "transformation" strategies (for Ranks 1 and 4). In addition, aligning with the report given by Munim et al. [24] and the NDEA model's preference for the transformation strategy, we substitute the values corresponding to the second and third ranks of the substitution strategy with those of the localization strategy. Introducing another mutation, we combine the preceding two mutations, resulting in four distinct scenarios: one rooted in Table II outcomes, while the other three stem from mutations.

The outcomes of the comprehensive evaluation are presented in Fig. 5. It is evident that, according to the second approach, the "single-use" strategy rises as the most suitable, whereas the transformation strategy prevails under the first approach. It is noteworthy that the disparity in consensus results stems from the differing perspectives of the approaches: The first approach aligns with a unified viewpoint on the scenarios, whereas the second approach seeks to mitigate the risks associated with the scenarios to a greater extent. Hence, SC managers are advised to select the optimal outcome based on their perspective and the internal as well as external environment of the organization.

To analyze the strategies for blockchain adoption using the four methods (SEDEA, SAW, TOPSIS, and NDEA) and compare them to the results of Munim et al. [24], according to Table III, we discuss the R-squared ($R^2$) and adjusted $R^2$ values. These values help us understand the extent to which the results from each method align with the results obtained by Munim et al. [24]. Here, $R^2$ represents the proportion of variance in the dependent variable (Y, results from Munim et al. [24]) that can be predicted from the independent variable (X, results from each method). A value closer to 1 indicates a strong predictive relationship. Adjusted $R^2$ modifies the $R^2$ value by accounting for the number of predictors in the model, providing a more accurate measure of goodness-of-fit, especially in models with multiple predictors. Like $R^2$, a value closer to 1 signifies a better model fit. The application of ML for similarity analysis and dual-perspective clustering is presented in the Supplementary Material (Section C) in detail to better understand interstrategy and method relationships.

## VI. Discussion

BT is being actively adopted by major O&G companies such as TotalEnergies, Aramco, and Shell to enhance SC coordination, optimize operations, and reduce environmental impact. The discussion highlights the adoption of BT by companies in the O&G industry, showcasing its practical applications. It explores strategies such as "single-use," "localization," "substitution," and "transformation," and presents decision-making models of EM, TOPSIS, SAW, and DEA to guide effective implementation. Challenges in BT adoption are acknowledged, with a proposed framework to optimize strategies in real-world conditions. The text also addresses research gaps, introducing a method to reconcile diverse evaluation outcomes, and suggests blockchain's potential to reduce environmental impact while enhancing coordination in SCs. The discussion demonstrates blockchain's practical relevance in the O&G sector, providing actionable insights and decision-making tools for industry professionals, with a focus on scenario-based and risk-based approaches.

Several prominent companies in the O&G industry are actively adopting BT in real-world applications. TotalEnergies Trading & Shipping, a prominent global player in commodities and oil product negotiations, is renowned for its expertise across key hubs spanning from Singapore to Houston, with an established presence in Geneva and Paris [43]. The imperative to accelerate digital trading transformation seamlessly aligns with the strategic vision of TotalEnergies Trading & Shipping. Serving as a catalyst for digitalization, the company actively engages in oil industry digitalization initiatives, including prominent blockchain consortia where it maintains significant shareholder positions. The operationalized Digital Lab is dedicated to pioneering innovative business solutions, augmenting portfolio management capabilities, enabling real-time vessel tracking, and identifying novel market opportunities. The integration of BT at oil fields and refineries has been instrumental in optimizing performance and streamlining the reconciliation of smart contracts with vendors. Aramco's proactive embrace of this technology has earned recognition from Forbes as an early adopter within the industry [44].

Shell's blockchain strategy is underpinned by three core pillars, leveraging the technology to overhaul processes, realize cost efficiencies, and standardize operations across the energy sector [45]. The company aims to discover new value opportunities in emerging markets by redefining end-to-end value chains through the adoption of BT. Emphasizing the importance of decentralized ecosystems, Shell actively supports public networks tailored for enterprise use and actively participates in a range of blockchain initiatives at varying levels of development. Collaborating with startups, joint ventures, and industry consortia, Shell explores novel business prospects. Significantly, Shell utilizes blockchain to verify the sustainability of hydrogen sources and has launched a blockchain-based solution to expand the use of sustainable aviation fuel. Through the trial of a decentralized digital passport system, Shell utilizes blockchain to improve tracking and safety management within SCs, striving to minimize costs and boost productivity. Established in 2017, the Shell Blockchain Centre of Excellence coordinates blockchain endeavors, shares knowledge, and establishes strategic alliances to navigate the evolving decentralized landscape. The center concentrates on achieving technological advantages, making informed technology decisions, and developing key capabilities. Comprising conceptual, legal, and technical experts alongside blockchain developers, the team collaborates closely with blockchain product teams embedded within Shell's businesses.

### A. Strategic Recommendations

The sophisticated intrinsic of the O&G industry, marked by sensitive data, diverse stakeholders, extensive paperwork, and transaction complexities, necessitates the swift adoption of BT in its SC operations. Preferred BT adoption strategies

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were to be based on a well-balanced improvement, multiple approaches of decision making already determined with respect to efficiency, practicality, and risk considerations, with optimization, statistics analysis, and ML concepts. To navigate this terrain effectively, SC managers must meticulously evaluate blockchain deployment strategies using comprehensive criteria and aggregate evaluation results to arrive at informed decisions. In this view, managers can incorporate multiple perspectives into their assessments using advanced decision-making methods, e.g., DEA for efficiency benchmarking, SAW for simple scoring, and TOPSIS for ranking alternatives based on the ideal-solution approach. In addition, the EM for weighting criteria guarantees that the key critical factors are prioritized without bias. Employing an aggregation method grounded in the scenario approach enables the integration of uncertainty and risk factors into the decision-making process, enhancing managerial insights and guiding strategic choices.

SC managers should thoroughly assess various blockchain deployment strategies, considering a wide range of factors such as coordination with SC partners, customer demands, market conditions, and information security requirements. It is essential to bring together evaluation findings into a cohesive summary that supports informed, data-driven decision-making. In this study, managers can derive rankings that stand the test of uncertainty and adjust to changing priorities through a consensus approach aimed at combining the outputs of these different models by scenario and risk. Given the complex and multiparty nature of OGSCs, fostering stronger collaboration among stakeholders is crucial. BT can significantly enhance transparency and streamline cooperation, leading to improved operational performance and increased trust throughout the network. Stakeholder cooperation requires the establishment of a shared understanding among them in evaluating and reaching a consensus on the results.

Strategic decisions made by SC managers should be closely aligned with the outcomes of these evaluations. By drawing on insights from blockchain strategy assessments, they can prioritize strategies that demonstrate high levels of efficiency and effectiveness, ultimately improving return on investment and supporting broader organizational objectives. The varying results obtained from different evaluation models highlight the importance of building consensus among stakeholders. Managers should encourage collaborative decision-making to reconcile differing viewpoints and develop a unified approach to blockchain implementation, which is key to gaining organizational support and ensuring successful execution. As the O&G industry continues to evolve, so too must its strategies. Managers need to establish mechanisms for ongoing monitoring and feedback to enable timely adjustments in response to changes in the market, advancements in technology, and shifting business priorities. The proposed framework inherently incorporates dynamic adaptation, enabling flexible adjustments to market fluctuations and organizational strategies. Dynamic adaptation means the framework and related parameters can be easily updated or recalibrated as circumstances change. Effective communication of evaluation results and recommendations is also vital for aligning the organization around blockchain initiatives. The proposed framework presented in Fig. 2 can also be implemented across different time periods and with various datasets. It means that the framework has been validated in the O&G sector in Norway, but its structured approach is general and can be adapted for other industries. Based on periodic evaluations, managers can optimize resource allocation and develop the necessary infrastructure.

In this study, besides the integrated multimodel evaluation framework, advanced ML techniques, such as K-means and hierarchical clustering, are applied in order to reveal subtle similarities and differences among BT adoption strategies and evaluation methods. Within the framework for similarity assessment and unsupervised clustering (including K-means and hierarchical clustering), an advantage is granted to managers in that clear visual groups can be established, distinguishing performance levels among strategies, making top performers, for instance, "single-use," easily recognizable along with moderate or weaker alternatives, and group methodologically related strategies, such as clustering SAW and TOPSIS results. This twofold-clustering analysis enables managers to both validate model outcomes and make more robust and nuanced decisions. In summary, the proposed multimodel decision-making framework provides substantial value to managers by improving strategic decision-making under uncertainty in the context of blockchain adoption. Significantly, the evaluation framework proposed here provides managers with a comprehensive, straightforward, easy-to-adapt evaluation of strategy as changes in the business environment occur scenario by scenario. This means that the use of multimodel decision analytics combined with advanced clustering methods enables higher clarity for the managers to distinguish similarities and contrasts among their blockchain strategies, even across varied perspectives for evaluation. This setup allows for movement in the data with reassessment as market conditions, risk profiles, or organizational priorities change. Hence, even though most insights in the work stem from the Norwegian O&G context, the modularity of the framework, combined with a data-driven and consensus-based approach, makes it generalizable and customizable to most industries and decision contexts. Table D.2 in the Supplementary Material summarizes key managerial implications.

### B. Operational Recommendations

The practical relevance of our findings was further validated through a case study in the Norwegian O&G sector. To provide more *operational* recommendations for managers based on the findings, it is essential to discuss how the proposed framework can be implemented in practice. The decision-making models utilized in this study offer unique advantages in evaluating blockchain implementation strategies, ensuring a structured and data-driven approach. Managers are recommended to take advantage of equations (A1) to (A5) to systematically determine the relative importance of each evaluation criterion, focusing on key aspects such as collaboration with SC partners, customer demands, market trends, and information security. The EM is especially well-suited for this purpose, as it allows for objective weighting based on the variation observed in the data.

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To assess different blockchain deployment strategies, a range of models should be employed. The SEDEA model helps identify the most efficient strategy while offering deeper insights into how alternatives compare in terms of performance. The SAW method provides an easy-to-understand initial evaluation through the summation of weighted performance scores. The TOPSIS method ranks alternatives based on their distance from the ideal solution, helping to pinpoint the most favorable options. Meanwhile, NDEA expands the analysis by evaluating efficiency across linked processes, offering a broader understanding of how strategies affect the entire network. Bringing together findings from these various models provides a more balanced view of possible outcomes. A scenario-based analysis using (1) to (4) effectively accounts for uncertainty and risk, enhancing the reliability of managerial decisions. Discrepancies between model results should be resolved through collaborative discussions and computations, ensuring agreement on the selected blockchain strategy and facilitating successful execution.

The analysis indicates that "single-use" and "localization" strategies consistently perform best across different evaluation models. These approaches should be prioritized for blockchain implementation, particularly in environments with higher risks. Balanced operational environments, particularly those prioritizing weighted scenario analysis, often make transformational strategies the most suitable option. In conclusion, the study shows that BT adoption in the Norwegian O&G industry is guided by strategies such as "single-use" and "localization," which are ranked highest for effectiveness using decision-making models such as SAW, TOPSIS, and SEDEA. These models help assess criteria such as market pressure, collaboration with SC partners, IT infrastructure, and operational costs. The study finds that SAW and TOPSIS align most closely with previous research, making them strong tools for strategy evaluation. The most important challenges include MP, NCS, LI, and CP. A consensus optimization model is proposed to improve decision-making compared to simpler methods such as the geometric mean. Norwegian O&G firms are advised to adopt a collaborative, risk-based, and scenario-focused approach when implementing BT.

## VII. Conclusion

Decision-making stands as a pivotal process wielded by SC managers, exerting a profound influence on the efficacy of SCM. Particularly in the contemporary landscape, the judicious utilization of emerging technologies such as blockchain within the SC domain assumes paramount importance. Among the various sectors, the O&G industry emerges as a valuable foundation for the adoption of BTs. In order to empower SC managers with the tools necessary for effective evaluation of blockchain deployment strategies within their SCs, the employment of decision-making models becomes imperative. Within the framework of this work, a collection of models utilizing principles such as ideal and anti-ideal points, network theory, and data-driven optimization was applied to systematically analyze these strategies. The outcomes yielded by divergent evaluation methodologies often confound SC managers. To mitigate this confusion, a consensus model contingent upon risk and scenario analysis was developed to integrate the evaluation results obtained from the Norwegian O&G industry. The findings delineated that the transformation strategy reigns supreme under the scenario-oriented approach, while under the most conservative risk containment approach, the "single-use" strategy rises as the most effective BT implementation strategy within the Norwegian O&G sector. Despite the promising results and comprehensive approach of this work, some limitations must be acknowledged.

The study primarily focused on the Norwegian O&G sector. While this industry presents a conducive environment for blockchain applications due to its complexity and scale, the findings may not be directly applicable to other industries without further investigation. Another limitation was the static nature of the analysis. It did not account for the dynamic nature of SCs. Future research might aim to investigate the proposed decision-making models across different industries. This would enhance the generalizability of the findings and recommendations. On the other hand, the inclusion of more decision-making models [48], [49] could provide a more comprehensive evaluation framework. Further exploration of uncertainty and risk factors in blockchain implementation could provide deeper insights. Future research could focus on the practical aspects of integrating blockchain with existing technologies and infrastructure within SCs. This would involve examining interoperability, scalability, and the necessary investments for successful implementation. By tackling these limitations and chasing the proposed research directions, future work can expand upon the current findings and provide more exhaustive, generalizable, and actionable insights for SC managers considering BT.

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